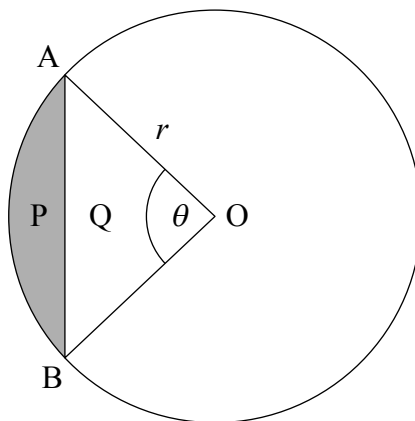


6. [Maximum mark: 6]

The following diagram shows a circle with centre O and radius r cm. Points A and B lie on the circle and $\angle AOB = \theta$ radians.

Sector OAB is divided into two regions, a shaded segment P and a triangle Q .



The area of the shaded segment P is 12.8 cm^2 .

The areas of P and Q are in the ratio $3:5$.

Find the value of r .

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